COURSE · SWE 4404: SPL-II

DENGUE ALERT

Community-Based Dengue Risk Reporting & Ward Management System

FINAL PROJECT PRESENTATION

GROUP 14

PROJECT TEAM

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PROJECT OVERVIEW

COMMUNITY HEALTH & SAFETY

Dengue Alert

A community-based dengue risk reporting and response system

Turns citizen observations into location-aware reports that local teams can review, assign, and track.

01 **Citizen reporting**
Photo + GPS evidence

02 **Ward-aware triage**
Spatial boundary lookup

03 **Coordinated response**
Councillor and inspector workflows



REAL APPLICATION INTERFACE

From a reported risk to a coordinated local response.

WORKFLOW

From a citizen report to a coordinated response

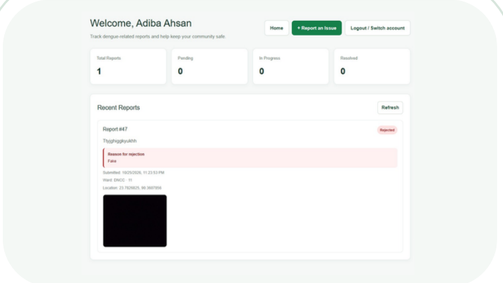
01 CITIZEN
Captures a photo and GPS location

02 BACKEND
Validates and maps the report to a ward

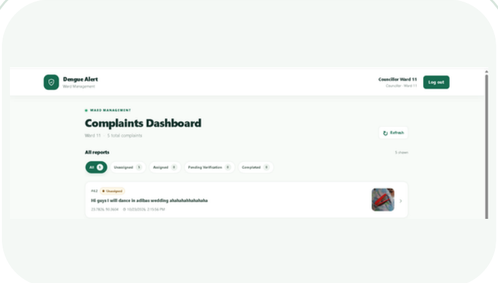
03 COUNCILLOR
Reviews complaints and assigns tasks

04 INSPECTOR
Completes field work and submits evidence

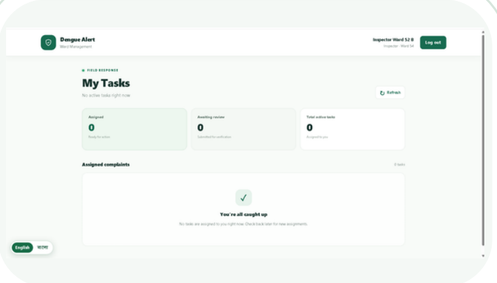
05 MASTER ADMIN
Manages staff and monitors hotspots



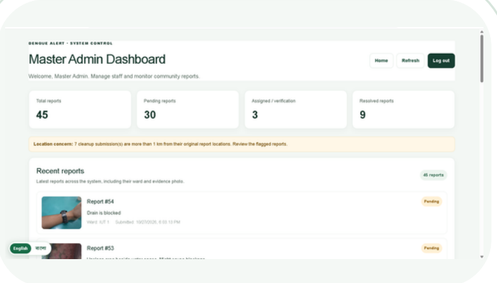
Citizen dashboard
Report tracking



Councillor dashboard
Complaint review



Inspector dashboard
Assigned tasks



Master Admin dashboard
System monitoring

Citizen proximity alerts can warn when an active hotspot is within the configured radius (default: 200 m).

PROJECT STRUCTURE

FRONTEND · React / JSX

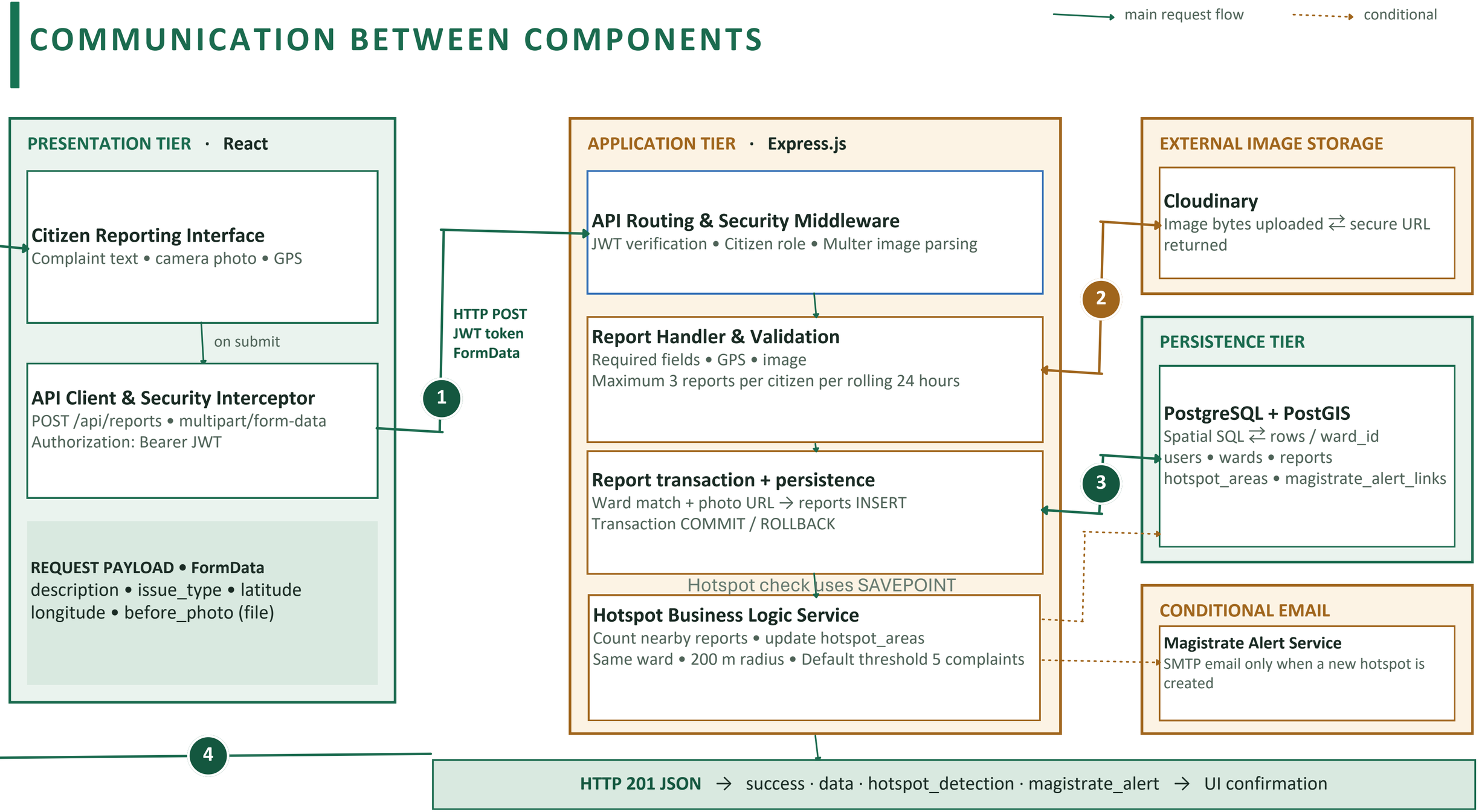
```
src/
├── App.js
├── axios.js
├── index.js
├── index.css
├── setupProxy.js
├── service-worker.js
├── serviceWorkerRegistration.js
├── context/
│   └── AuthContext.js
├── i18n/
│   ├── LanguageProvider.jsx
│   ├── bn.json
│   └── language.css
└── components/
    ├── BeforeAfterComparison.jsx
    ├── CleanupUpload.jsx
    ├── ConfirmDialog.jsx
    ├── GoogleIdentityButton.jsx
    ├── HotspotProximityMonitor.jsx
    ├── HotspotProximityMonitor.css
    ├── InspectorAssignmentCard.jsx
    ├── LiveCameraCapture.jsx
    ├── MapLocationView.jsx
    ├── Navbar.jsx
    ├── PasswordVisibilityToggle.jsx
    └── StatusBadge.jsx
```

```
src/pages/
├── Dashboard.js
├── MasterAdminPanel.js
├── LandingPage.jsx
├── LandingPage.css
├── ReportIssue.jsx
├── ReportIssue.css
├── MagistrateHotspotDetails.jsx
├── MagistrateHotspotDetails.css
├── ComplaintDetailTheme.css
├── ThemePages.css
├── auth/
│   ├── Login.js
│   ├── Register.js
│   └── VerifyEmail.jsx
├── councillor/
│   ├── ComplaintDetails.jsx
│   └── CouncillorDashboard.jsx
├── inspector/
│   ├── InspectorComplaintPage.jsx
│   └── InspectorDashboard.jsx
└── admin/
    ├── MasterAdminDashboard.jsx
    └── MasterAdminDashboard.css
```

BACKEND · Express / Node.js

```
SPL-II-23-c3/
├── server.js
├── seedwards.js
├── dhaka_wards.geojson
├── package.json
├── package-lock.json
├── README.md
├── .gitignore
├── config/
│   ├── db.js
│   └── cloudinary.js
├── middleware/
│   ├── authmiddleware.js
│   ├── rolemiddleware.js
│   └── upload.js
├── routes/
│   ├── userRoutes.js
│   ├── reportRoutes.js
│   ├── inspectorRoutes.js
│   ├── hotspotRoutes.js
│   └── magistrateAlertRoutes.js
├── services/
│   ├── hotspotService.js
│   ├── proximityAlertService.js
│   ├── magistrateAlertService.js
│   ├── verificationMail.js
│   └── passwordPolicy.js
└── tests/
    ├── hotspotService.test.js
    ├── proximityAlertService.test.js
    └── magistrateAlertService.test.js
```

COMMUNICATION BETWEEN COMPONENTS



DEMONSTRATION

GITHUB REPOSITORY: SPL-II-23

Thank You